Title: Identifying trends in reporting on the ethical treatment of insects in research

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Abstract

Transparent reporting on the ethical treatment of research animals (e.g., implementation of the 3Rs, replace, reduce, and refine) is recommended when publishing in the peer-reviewed literature. This is meant to foster public trust, safeguard animal welfare, and generate reproducible science. However, entomologists are not expected to engage in such reporting, as their research is not subject to legislated ethical review. Recently, however, entomologists have reported increased concern about the ethical treatment of insects in research, and associated reproducibility and public trust issues. To what degree are these increasing concerns reflected in changes in practices? We surveyed 15 high-impact journals that publish on insects over 20 years to collect data on reporting related to the ethical treatment of insects in research, including animal reduction methods, analgesics/anesthesia statements, and information regarding sacrifice. Out of 1359 sampled papers, no studies reported any methods to reduce animal use. Over 20 years, we found an increase in the proportion of papers reporting insect death and a decrease in the papers reporting significant invasive handling. 84% of papers with significant animal handling or death did not report the use of any anesthetics. We also found an increase in animal-treatment-specific ethics statements (from 0% to 8%), largely driven by the journal *Animal Behaviour*. We end by 1) making recommendations for entomologists looking to improve their reporting practices and 2) providing tools to improve transparent reporting of information related to the ethical treatment of insects in research.

Introduction

The ethical treatment of animals used in research is a cornerstone of generating reproducible science, safeguarding animal welfare, and fostering public trust. Transparent reporting on the practices used during animal experiments is thus recommended when publishing peer-reviewed literature (Sert et al., 2020). Historically, however, similar norms have not been considered best practice for animals excluded from legislated ethical oversight (Freelance, 2019; Harvey-Clark, 2011), such as insects [(Eisemann et al., 1984), but see (Gibbons et al., 2022)]. In these cases, disciplines like entomology have developed permissive research and reporting norms for animal use relative to those that are standard for other animals. Where these permissive norms create risks for 1) public trust, 2) animal welfare, or 3) reproducible science, entomologists may wish to review their norms and establish new best practices for the field.

There are reasons to think that permissive norms indeed create such risks. For instance, recent research shows that public trust in invertebrate research declines without ethical oversight (Brunt et al., 2022), suggesting that entomologists may have just as much need to transparently report on ethical practices used in research as vertebrate biologists. Further, public opinion on insect sentience (i.e., the capacity for valenced states like pain) may be shifting (Drinkwater et al., 2019), at least in Western nations. 47.2% of American adults believed insects used as livestock feed could feel pain, with only 18.9% uncertain; further, younger respondents were more likely to be concerned about the ethical treatment of insects (Fukuda et al., 2023). Numbers may be even higher for more charismatic animals commonly used in the lab, such as bees: a survey of American adults found differences in the public's belief in the plausibility of insect pain was highest for bees (65% yes, 27% unsure), compared to ants

61 (56% yes, 32% unsure) or termites (52% yes, 35% unsure; Dullaghan et al., 2021). Recent examples of
62 public outcry about lethal insect collecting studies (e.g., BBC Radio 4, 2018) suggests that the public is
63 becoming more concerned about ethical treatment of insect research subjects, which matches growing
64 attention to this issue by researchers as well (Barrett et al., 2023; Fischer & Larson, 2019; Lövei et al.,
65 2023).

66 Although there is no scientific consensus on insect sentience, a recent review (Gibbons et al.,
67 2022) suggests there is enough neurobiological and behavioural evidence to warrant precautionary
68 approaches towards their welfare in research until more data can be collected (Andrews et al., 2024;
69 Crump et al., 2023; Cuff et al., 2023; Fischer & Larson, 2019; Klobučar & Fisher, 2023; Sandall &
70 Fischer, 2019). Indeed, such a precautionary approach has even been recommended by professional
71 entomologists who are skeptical about insect sentience (Eisemann et al., 1984). Reporting on ethical
72 animal use practices is itself a precautionary measure, encouraging animal researchers to articulate and
73 refine their practices to reduce avoidable harms (Nawroth & Krause, 2022).

74 Finally, reporting on ethical animal use is a critical part of reproducible science wherever either
75 affective states or physiological/behavioural changes due to stressors may be expected to confound or
76 otherwise alter study results (see examples from the vertebrate literature: Cait et al., 2022; Loss et al.,
77 2021; Soulsbury et al., 2020). Just as in vertebrates, studies in insects suggest that many welfare-
78 relevant practices can influence study results. For example, some methods of anesthesia (cold, CO_2)
79 can impact behaviour for days after the anesthetic was applied in *Drosophila melanogaster,* while
80 others (halogenated ethers) do not have the same behavioural effects (MacMillan et al., 2017). A lack
81 of reporting on the anaesthetics used, the duration, and the dosage applied can thus represent a

reproducibility concern in the literature that better reporting on animal welfare would simultaneously address (Carbone & Austin, 2016).

A recent survey of entomologists suggests there is growing concern about insect welfare based on the considerations just surveyed—i.e., preserving public trust, the intrinsic value of animal welfare itself, and generating reproducible science (Barrett et al., 2024). However, it is unclear whether and how this increase in concern is reflected in practice. One valuable public metric is increased reporting of welfare-relevant information in entomological publications. To determine trends in historical reporting of the ethical treatment of insects, we survey five papers per year from the past 20 years (2003-2022) that use live insects from 15 peer-reviewed journals with a range of impact factors, subdisciplinary breadth, and taxonomic focus.

We report on trends in the data related to two components of the standard 3Rs framework for ethical animal use in research: reduction (using fewer animals when possible) and refinement (minimizing harm to living animals when possible; Russell & Burch, 1959). This framework has recently been recommended by several authors as a best practice for use with invertebrates (Crump et al., 2023; Montero-Castaño et al., 2022). We thus looked for reports that authors were reducing animal use to the minimum number required (e.g., through the use of statistical methods for determining appropriate power), or refining their practices via the use of anaesthetics/analgesics prior to significant handling or non-instantaneous sacrifice. Further, we discuss the prevalence of explicitly-labeled ethics statements. Finally, we make recommendations for how entomologists could improve transparent reporting of information related to the ethical treatment of their research subjects in the

peer-reviewed literature, with the goal of addressing concerns about public trust, welfare, and reproducibility.

Methods

Journal and Article Selection

We used the Scopus CiteScore rankings, as of January 2023, to determine the top 40 entomology journals. We excluded any journals from the list that, by name, focus mostly on review papers, non-animals (e.g., plants, fungi, etc.), or non-insect animals (e.g., ticks, etc.). From the remaining 26 journals, we selected 9 that represented a broad swath of applied and fundamental entomological disciplines, to capture a range of methods and sub-disciplinary practices. Applied insect-focused journals included: *Journal of Pest Science, Insect Conservation and Diversity, Medical and Veterinary Entomology*, and *Journal of Insects as Food and Feed.* Fundamental insect-focused journals included: *Insect Biochemistry and Molecular Biology, Insect Science, Journal of Insect Physiology, Ecological Entomology*, and *Arthropod Structure and Development.* We selected an additional six, high-impact journals that publish fundamental papers from a variety of subdisciplines, but that are not taxonomically focused on insects despite frequently featuring insect work: *Nature, PNAS, Journal of Experimental Biology, Animal Behavior, eLife*, and *Proceedings of the Royal Society B.* These journals were selected to better understand if publishing alongside vertebrate articles would shape entomological ethics reporting practices.

Five articles were randomly selected from each journal for each year they were in operation between 2003 and 2022 (n = 1,359). Articles that did not report primary research on insect specimens

were not used and were replaced by another randomly selected article from that journal in that year, unless no additional articles that met these criteria could be found.

Article Assessment and Searches

The methods of each article were read and assessed to determine 1) if any insects were significantly handled during the experiment (e.g., restraint, dissection) and 2) if any insects died during the experiment (e.g., whole-body sampling for DNA) or as a direct result of experimental procedures (e.g., electrophysiology). Other papers likely involved insect death as part of their ongoing experimental procedure but animal sacrifice may not have been an explicit part of their protocol (e.g., rearing of fruit flies for behavioural investigations). These were not recorded as having involved animal sacrifice. The methods of each paper were also searched for reporting of any *a priori* statistical analyses for assessing the number of animals needed to complete an experiment (e.g., power analyses).

Further, articles were searched automatically for the following keywords: 'anesth*', 'anaesth*', 'analg*', 'welfare', 'ethic*', 'handling', 'manipulation', 'cold', 'freeze*', 'froze*', 'ice', 'liquid (nitrogen)', 'carbon (dioxide)', 'CO2', 'ether', '*flurane', 'euthan*', 'sacrific*', 'ethanol', 'morphine', 'opioid', 'GABA', 'gaba*', 'ɣ-aminobutyric acid', 'potassium chloride', 'NSAID', 'antidepressant'. When these terms resulted in a 'hit', the area of the paper containing the search term was read to determine if the hit was welfare relevant (e.g., a sacrifice or analgesia/anaesthesia statement, or something else). For welfare-relevant hits, the method of anesthesia/sacrifice was recorded, the location of the hit within the paper (methods, or elsewhere), and if the hit was part of an explicit 'ethics' statement or only in the general methodology. Only the main materials (i.e., not supplemental files) of publications were searched for these terms.

Statistical Analyses

All statistical analyses were conducted using R v. 4.4.0 (R Core Team, 2022). Linear regressions, Pearson-product moment correlations, *t*-tests, ANOVAs and Tukey Honest Significant Difference tests were implemented using the R base package. Alpha was set to 0.05.

Results

Throughout the results, 'ethics statement' refers to any ethics statement appearing in a paper if that statement was welfare-relevant (e.g., statements about permitting, or that say 'no ethics approval required for insects', were not excluded). 'Anesthetic/analgesic statement' refers to any mention in the paper of an anesthetic or analgesic being employed during the study, whether that information is reported for ethical or methodological reasons. 'Sacrifice statement' refers to any mention in the paper of an animal being sacrificed, whether that information is reported for ethical or methodological reasons.

Description of Sacrifice and Anesthetic Usage in Papers

No papers in our sample reported the use of any *a priori* methods (e.g. power analyses) for reducing the number of animals used in the experiment.

A mean (± standard deviation) of 77.17 ± 4.79% of papers across all years contained insect death during or resulting from studies (range: 63.33 - 82.35%; Table 1). 59.90% ± 7.18% of papers contained insect death with no significant handling (48.33 - 70.77%) while 17.11% ± 5.62% of papers contained significant handling prior to death (range: 7.69 - 27.12%). 10.27% ± 5.04% of papers contained significant handling not followed by death (range: 4.00 - 23.33%). Finally, 12.48% ± 3.25%

of papers contained neither significant handling nor death resulting from the study (range: 5.08 - 18.33%).

Table 1: Number of papers sampled and their respective counts and proportions of animal sacrifice and invasive handling per year.

Publication year	Papers sampled	All papers containing sacrifice		Papers containing sacrifice without invasive handling		Papers containing both invasive handling and sacrifice		Papers containing invasive handling without sacrifice		Papers containing neither sacrifice or invasive handling	
	n	n	%	n	%	n	%	n	%	n	%
2003	59	48	81.36	32	54.24	16	27.12	8	13.56	3	5.08
2004	60	42	70.00	29	48.33	13	21.67	11	18.33	7	11.67
2005	60	38	63.33	29	48.33	9	15	14	23.33	8	13.33
2006	61	45	73.77	37	60.66	8	13.11	11	18.03	5	8.2
2007	59	46	77.97	41	69.49	5	8.47	6	10.17	7	11.86
2008	65	51	78.46	46	70.77	5	7.69	4	6.15	10	15.38
2009	65	50	76.92	37	56.92	13	20	9	13.85	6	9.23
2010	69	54	78.26	45	65.22	9	13.04	7	10.14	7	10.14
2011	60	43	71.67	36	60.00	6	10	6	10	11	18.33
2012	62	51	82.26	41	66.13	10	16.13	4	6.45	7	11.29
2013	68	56	82.35	46	67.65	10	14.71	5	7.35	7	10.29
2014	68	55	80.88	44	64.71	10	14.71	4	5.88	9	13.24
2015	77	56	72.73	39	50.65	17	22.08	10	12.99	11	14.29
2016	75	60	80	50	66.67	10	13.33	7	9.33	8	10.67
2017	75	59	78.67	45	60.00	14	18.67	3	4	13	17.33
2018	75	58	77.33	38	50.67	20	26.67	5	6.67	12	16
2019	74	57	77.03	41	55.41	16	21.62	7	9.46	10	13.51
2020	73	57	78.08	40	54.79	17	23.29	4	5.48	12	16.44
2021	75	61	81.33	50	66.67	11	14.67	5	6.67	9	12
2022	79	64	81.01	48	60.76	16	20.25	6	7.59	9	11.39

The most common anesthetics/analgesics (Table 2) used prior to significant handling or death (and found as a result of our search terms) were freezing/chilling (n = 125 reports) and carbon dioxide

(n = 80). Less commonly reported methods were chloroform (n = 1), nitrogen/nitrogen monoxide (n = 2), ether (n = 3), isoflurane (n = 1). 84.16% of papers that reported significant handling or death did not report the use of any anesthetics/analgesics prior to the event (Table 2).

Table 2: List of anaesthetics reported in papers with invasive handling or insect sacrifice. Not stated category refers to those papers with invasive handling or insect sacrifice but no reported anaesthetic or analgesic information.

Anaesthetic	n	%
chloroform	1	0.08
isoflurane	1	0.08
nitrogen	1	0.08
nitrogen monoxide	1	0.08
ether	3	0.25
freezing	20	1.68
CO_2	80	6.74
chilling	105	8.85
not stated	999	84.16

When sacrifice methods were captured by our automated methods, cold/freezing/LN2/dry ice (n = 120) and dissection (n = 48) were the most commonly reported (Table 3). Less commonly reported methods were immersion in ethanol/solvent/fixative (n = 24) and homogenisation (n = 15), among others. 79 studies that our automated methods highlighted as containing an animal sacrifice method did not report which method was used to kill their animals (38.05%).

Table 3: Sacrifice methods reported in papers extracted by our automated process. Not stated category refers to those papers identified as having an insect die as a result of their methods but that did not describe the process that resulted in death.

Sacrifice method	n	%
boiled	1	0.49
cold	1	0.49
diethyl ether	1	0.49
ether	1	0.49
glycerol	1	0.49

irradiated	1	0.49
CO_2	2	0.97
pesticide	2	0.97
dry ice	3	1.46
fixative	3	1.46
solvent	6	2.91
ethanol	11	5.34
homogenized	15	7.28
liquid nitrogen	33	16.02
dissected	48	23.3
not stated	79	38.35
frozen	83	40.29

1164 total papers were manually identified as containing insect death, with 873 papers that contained insect death but were not recognized by our automated methods. Of the papers where the sacrifice method was captured manually (Table 4), the most common methods were dissection (n = 304) or cold/freezing/LN2/dry ice (n = 137). The largest contingent of papers (Table 4; n = 307), did not explicitly state any sacrifice method but on examination of the methods, death could be inferred with certainty. The most common experimental procedures from which insect death could be inferred (Supplemental Table 1) were nucleic acid extraction (n = 119) and electrophysiology (n = 28).

Table 4: Sacrifice methods reported in papers extracted by manual searching.

Sacrifice method	n	%
dry ice	1	0.1
microwave	1	0.1
CO2	2	0.19
cold	2	0.19
irradiated	2	0.19
starved	2	0.19
Bt toxin	3	0.29
heat treatment	3	0.29
drowned	5	0.48

boiled	6	0.57
crushed	7	0.67
dessicated	8	0.76
infection	11	1.05
glycol	12	1.14
solvent	12	1.14
predation-parasitism	14	1.33
exsanguinated	18	1.71
liquid nitrogen	20	1.90
decapitated	25	2.38
trap	27	2.57
homogenized	30	2.85
fixative	32	3.04
experimental death*	56	5.33
pesticide	58	5.52
ethanol	78	7.42
frozen	115	10.94
dissected	304	28.92
not stated**	307	29.21

*The "experimental death" category refers to those papers were insects died through the course of rearing or during development.

*** The "not stated" category refers to those papers identified as having an insect die as a result of their methods but did not describe the process that resulted in death.

Of all the papers reporting dissection as a sacrifice method, 52 (17.11%) reported the use of an anaesthetic or analgesic prior to the dissection, with chilling/freezing (n = 33) and carbon dioxide (n = 18) the most commonly employed methods (Supplemental Table 2).

Trends in Reporting Over Time

The percentage of studies that contained any insect death increased from 2003 to 2022 (Figure 1; linear regression; $t_{20,18} = 2.33$, $p = 0.031$, $R^2 = 0.23$) while the percentage of studies that contained just invasive handling decreased (linear regression; $t_{20,18} = 4.00$, $P = 0.001$, $R^2 = 0.44$). There were no other significant changes over time in the use of invasive handling only (not followed by death), death

only (not preceded by invasive handling), or neither handling nor death (linear regression; $t_{20,18} < 2.06$, $p > 0.05$).

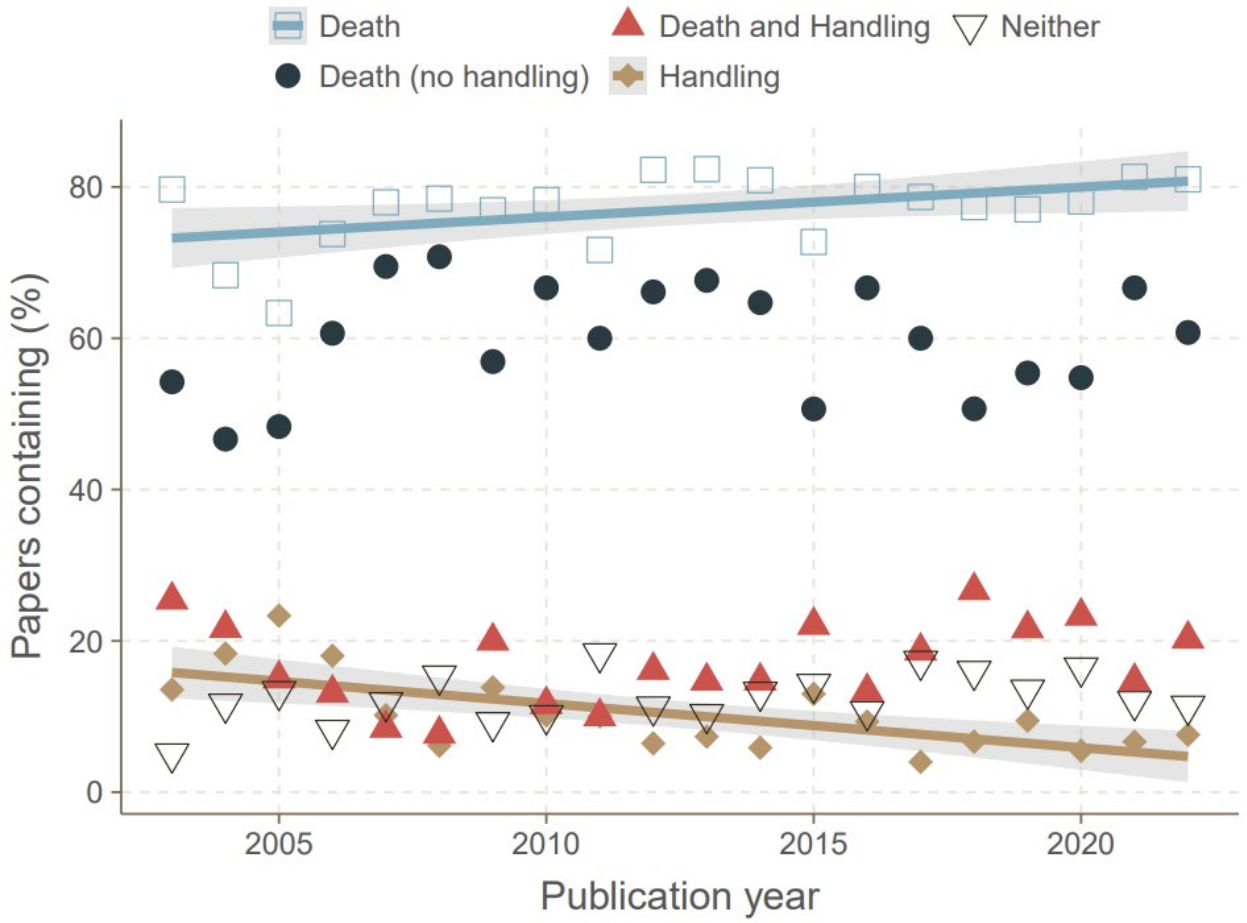


Figure 1. The percentage of studies that resulted in insect death increased over time. There was a significant increase in the proportion of studies containing any form of insect sacrifice over time (linear regression; $t_{20,18} = 2.33$, $p = 0.031$, $R^2 = 0.23$). There was a significant decrease in the proportion of studies reporting just invasive handling without sacrifice over time (linear regression; $t_{20,18} = 4.00$, $p = 0.001$, $R^2 = 0.44$). There was no change in the percentage of studies containing death only (no handling), both death and significant handling, or neither death nor significant handling (linear regression; $t_{20,18} < 1.19$, $p > 0.05$).

The percentage of studies that contained an ethics statement increased over time from 0.00% in 2003 to 7.59 % in 2022 (maximum = 9.59% in 2020) (Figure 2; linear regression, $t_{20,18} = 4.29$, $p < 0.001$, $R^2 = 0.48$). There was no significant relationship over time in the percentage of studies that

contained dissection with no accompanying anaesthesia or analgesia statement (linear regression; $t_{20,18}$ = 1.62, p = 0.12)

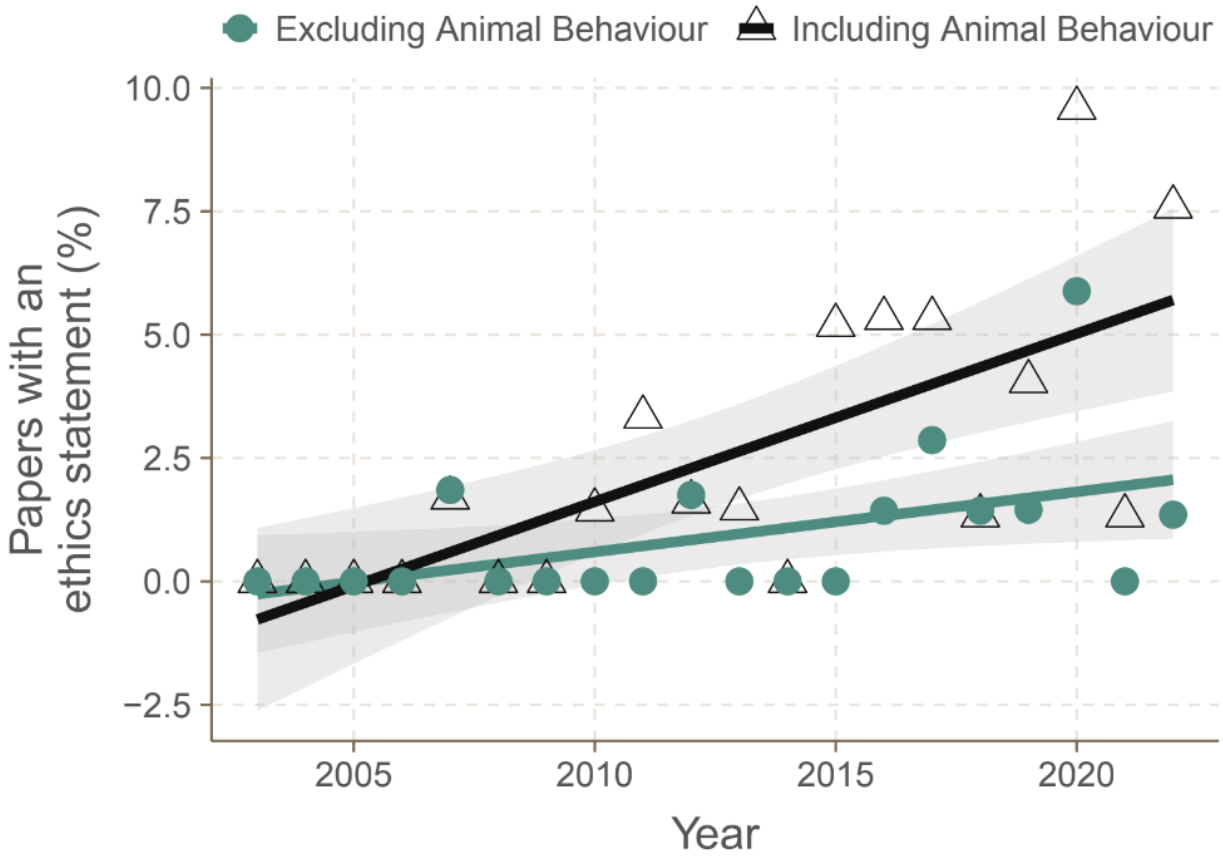


Figure 2. The percentage of studies that contained an ethics statement increased over time, largely driven by *Animal Behaviour*. There was a significant increase in the proportion of studies containing an ethics statement over time (black line; linear regression, $t_{20,18}$ = 4.29, p < 0.001, R^2 = 0.48). Much of this increase in the use of ethics statements was driven by publications in *Animal Behaviour,* however even with this journal excluded there was still a significant increase from 0.00% in 2003 to 1.35% in 2022 in the proportion of studies containing an ethics statement in the other 14 journals (green line; linear regression, $t_{20,18}$ = 3.83, p = 0.001, R^2 = 0.42).

Trends in Reporting by Journal

Comparing prevalence of ethics statements among journals (Figure 3) indicated that the aforementioned increase in ethics statements over time was largely driven by a single publication:

Animal Behaviour. However, even when excluding this journal from the analysis, there was still a slight but significant increase in the use of ethics statements across the other journals from 0.00% in 2003 to 1.35% by 2022 (Figure 2; linear regression, $t_{20,18} = 3.83$, $p = 0.001$, $R^2 = 0.42$).

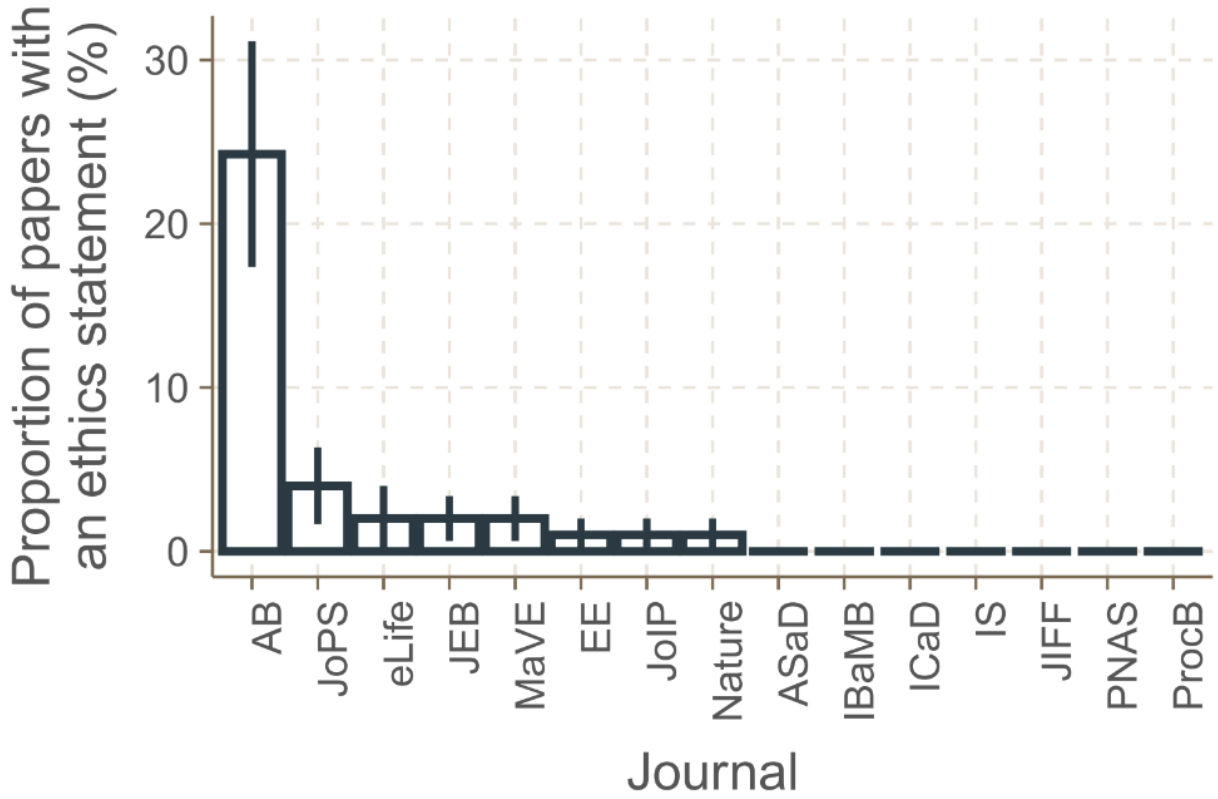


Figure 3. The journal *Animal Behaviour* has a greater proportion of papers with an ethics statement than the other journals combined. (Mean ± standard error). None of the other 14 journals, excluding *Animal Behaviour*, have a significantly different proportion of papers with ethics statements from each other. Journal abbreviations: ASaD - *Arthropod Structure and Development*; EE - *Ecological Entomology*; IBaMB - *Insect Biochemistry and Molecular Biology*; IS - *Insect Science*; JEB - *Journal of Experimental Biology*, JoIP - *Journal of Insect Physiology*; JoPS - *Journal of Pest Science*; MaVE - Medical and Veterinary Entomology; *Nature* - Nature; PNAS - *Proceedings of the National Academy of Sciences*; ProcB: *Proceedings of the Royal Society B*; ICaD - *Insect Conservation and Diversity*; JIFF - *Journal of Insects as Food and Feed*; eLife - *eLife*.

Animal Behaviour is the only journal in our sample to currently require an ethics statement, even for studies on invertebrates. We therefore first tested for any significant differences in the

proportion of papers with an ethics statement among the 14 other journals, where no ethics statement was required when working with invertebrates. There was no significant difference among them (ANOVA; $F_{13,253} = 1.29$, $p = 0.22$). We then collapsed them into a single factor, in order to compare journals where no ethics statement is required for studies with invertebrates to journals where this is required - in this case, only *Animal Behaviour*. *Animal Behaviour* had a significantly higher mean proportion of ethics statements than the journals that did not require an ethics statement when working with invertebrates (Figure 4; $t_{19,273} = 3.38$, p-value < 0.01).

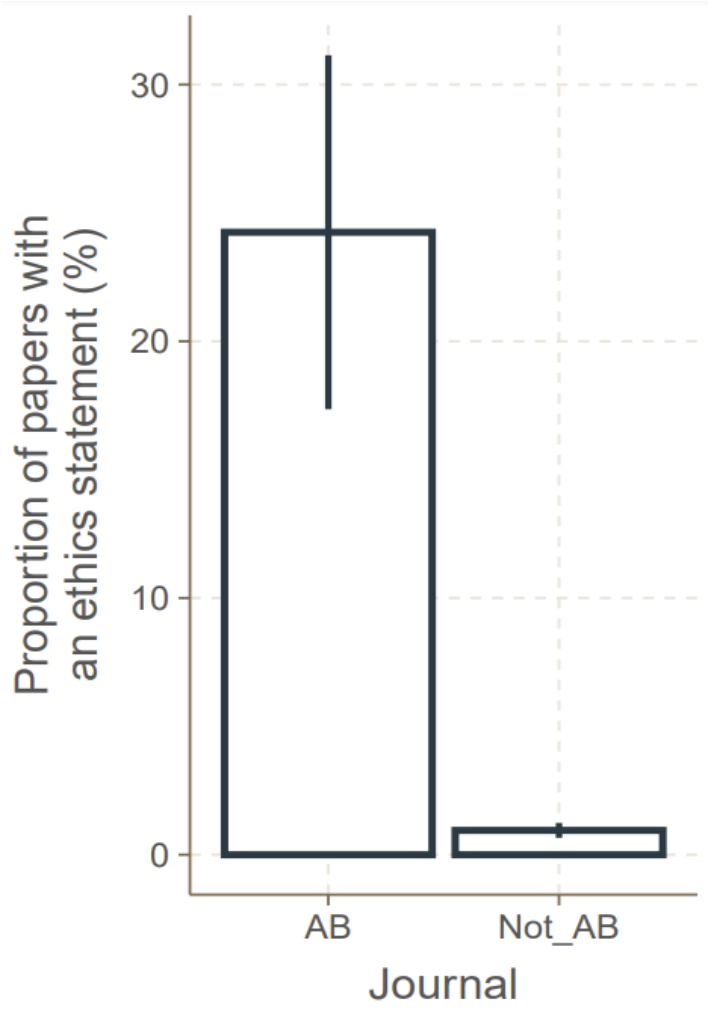


Figure 4. The journal that requires an ethics statement for studies that use invertebrates (*Animal Behaviour*) has a higher mean proportion of papers with an ethics statement compared to journals that do not require an ethics statement for studies that use invertebrates. *Animal Behaviour* had a significantly higher mean percentage of ethics statements (mean ± SE; 24.2 ± 6.89) than the journals that did not require an ethics statement (mean ± SE; 0.95 ± 0.29) when working with invertebrates, combined ($t_{19,273}$ = 3.38, p-value < 0.01).

Six journals (*Animal Behaviour, eLife, Journal of Experimental Biology, Nature, Proceedings of National Academy of Sciences, Proceedings of the Royal Society B*) from our sample publish papers investigating both invertebrates and vertebrates, while the remaining nine (*Arthropod Structure and Development, Ecological Entomology, Insect Biochemistry and Molecular Biology, Insect Science, Journal of Insect Physiology, Journal of Pest Science, Medical and Veterinary Entomology, Insect Conservation and Diversity, Journal of Insects as Food and Feed*) publish almost exclusively invertebrate research. Journals that also publish vertebrate research had a significantly higher proportion of papers reporting an anaesthesia or analgesia statement compared with journals that publish only invertebrate research (Figure 5; *t*-test, $t_{40,38,}$ 3.19, p < 0.003). Journals that publish vertebrate research also had a significantly higher proportion of papers reporting a sacrifice statement compared with journals that publish only invertebrate research (Figure 5; *t*-test, $t_{40,38}$ = 3.05, p < 0.005).

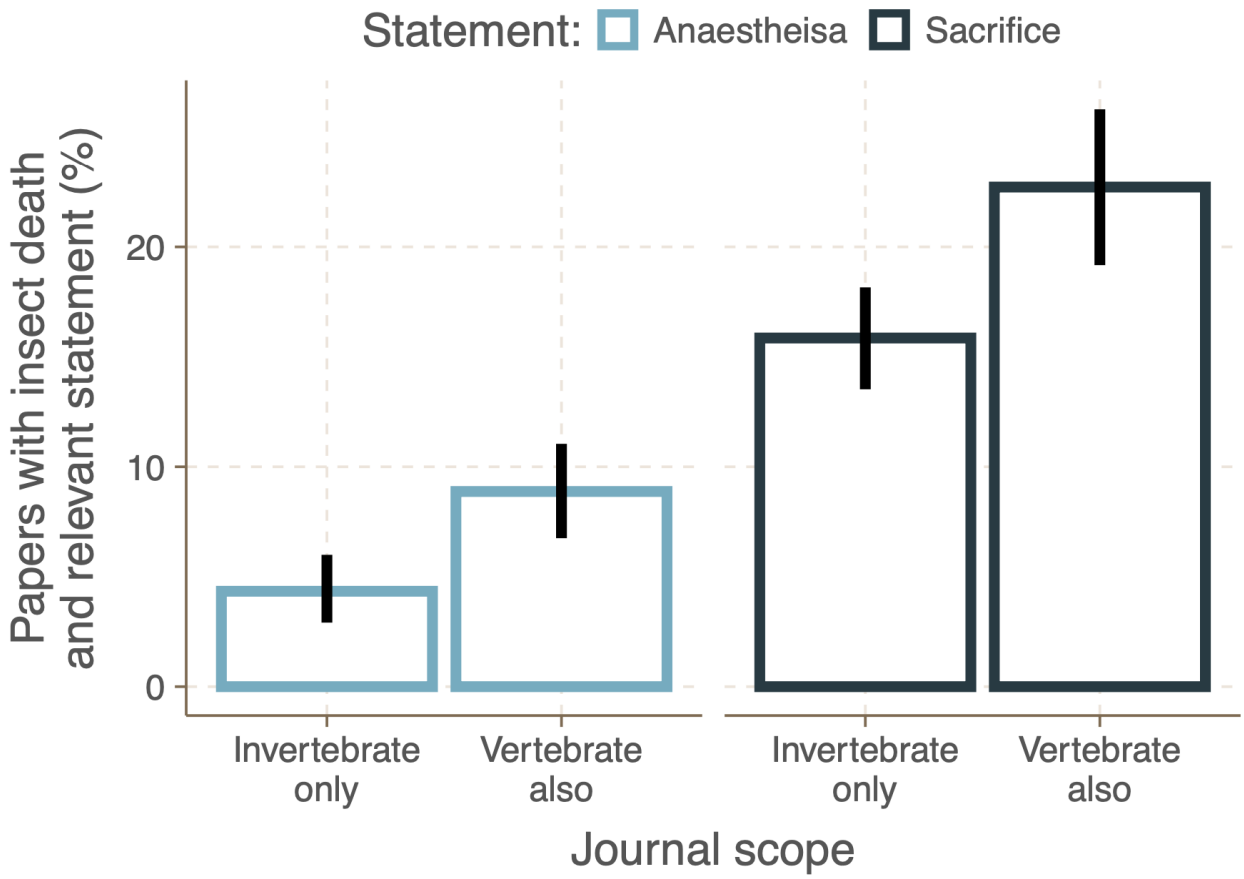


Figure 5. The percentage of papers reporting relevant anaesthesia or sacrifice statements by journal taxonomic scope. (blue) Journals that publish vertebrate research as well as invertebrate research had a significantly higher proportion of papers reporting an anaesthesia or analgesia statement compared with journals that publish only invertebrate research (*t*-test, $t_{40,3} = 3.19$, $p < 0.003$). (black) Journals that publish vertebrate research also had a significantly higher proportion of papers reporting a sacrifice statement compared with journals that publish only invertebrate research (*t*-test, $t_{40,38} = 3.05$, $p < 0.005$).

Trends in Reporting By Taxa

To facilitate robust statistical analysis, the seven most common orders (Supplemental Figure 1; Diptera, Hymenoptera, Coleoptera, Lepidoptera, Hemiptera, Orthoptera, Blattodea) within our sample were selected for analysis; all other taxa were collapsed into a single category ("Other Orders"), creating a total of eight categories. The seven most common orders accounted for 94% of the instances within the data.

There was a significant difference in the proportion of anaesthesia statements prior to invasive handling among orders (Figure 6A; ANOVA; $F_{7,8} = 6.36$, $p < 0.0001$). Investigations using Diptera contained a significantly greater proportion of relevant anaesthesia statements than investigations using Coleoptera (Tukey HSD; $p = 0.02$), Lepidoptera (Tukey HSD; $p = 0.001$), Orthoptera (Tukey HSD; $p < 0.0001$) and Other Orders (Tukey HSD; $p = 0.001$). Investigations using Hymenoptera had a greater proportion of relevant anaesthetic statements than those using Orthoptera (Tukey HSD; $p = 0.04$).

We found no significant difference in the proportion of sacrifice statements (Figure 6B; ANOVA; $F_{7,8} = 0.21$, $p = 0.98$), or the proportion of anaesthesia statements (Figure 6C; ANOVA; $F_{7,8} = 0.86$, $p = 0.54$), prior to death among orders.

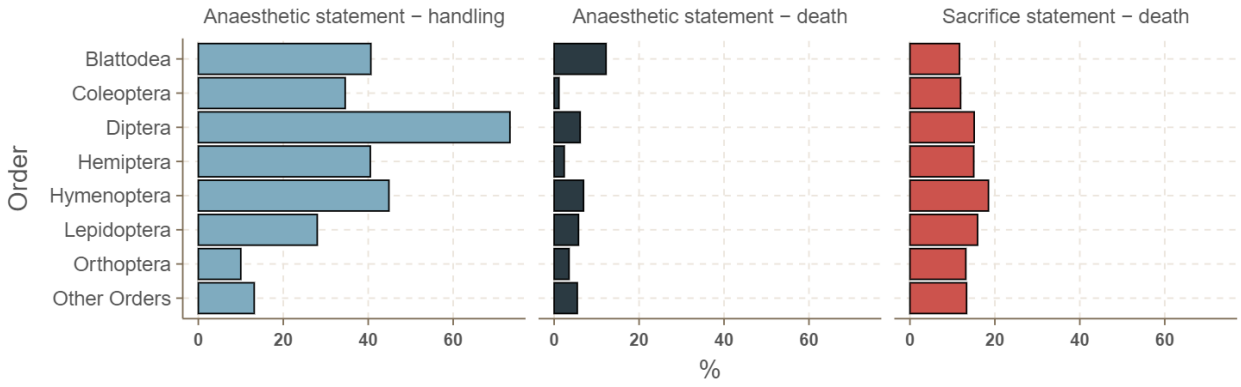


Figure 6. Percentage of studies with an anaesthetic or sacrifice statement prior to handling or death by order. Left) Percentage of studies with an anaesthetic statement prior to handling varied by order. Letters indicate statistically significant differences between orders. Middle) There was no significant difference in the percentage of studies with an anaesthetic statement prior to death among orders. Right) There was no significant difference in the percentage of studies with a sacrifice statement prior to death among orders.

Discussion

We sampled 1359 papers that used live insects from 15 journals across a 20 year period to ascertain trends in reporting on *a priori* power analyses, anesthetic use, and sacrifice methods across the discipline. Our study was only designed to look for specific indicators of 'reduction' and 'refinement' in studies where live animals were used (two of the 3Rs; Russell & Burch, 1959); we did not attempt to determine if there were opportunities in any studies to have replaced live animals with less-sentient alternatives (such as replacing live animals with museum specimens or performing studies on cells instead of whole organisms).

Reporting on animal reduction via a priori sample size estimation

No paper in our sample reported the use of *a priori* methods for determining sample sizes, a process that is recommended to reduce the number of animals used to the minimum necessary for a study to have appropriate power (Lakens, 2022; C. Perl, 2023). This practice has been shown to be essential for producing valid scientific conclusions as well as promoting animal welfare (Button et al., 2013; C. D. Perl & Fischer, 2024; Richter, 2024). Estimating sample sizes is a more common practice in vertebrate research, as ethical review boards can serve as a point of intervention during study design (Papalouka et al., 2023). However, recent studies suggest that methods for determining power or precision often are not used appropriately or transparently reported (Charles et al., 2009; Dumas-Mallet et al., 2017; Rutterford et al., 2015; Thibault et al., 2024). This can affect the reproducibility and validity of study results, ultimately contributing to translatability and generalizability issues (Spanagel, 2022). Increasing the correct use and transparent reporting (Sert et al., 2020; Spanagel,

2022) of these methods is an important part of scientific reproducibility and the growing movement towards open research practices (Cuff et al., 2023; Mellor et al., 2016).

Reporting on methods of sacrifice, anaesthetics, and analgesics

When considering refinements, we looked for reporting that demonstrated anaesthetics were being used prior to invasive handling or death. The majority of papers with invasive handling or death did not report the prior use of analgesics/anaesthesics (84%), despite the fact that papers reporting invasive handling and/or death were the vast majority of papers in our sample (88%). Further, when the use of anaesthetics was reported, freezing/chilling was the most frequently reported method (59%). However, invertebrate veterinarians have consistently stated that cold is not considered an appropriate anesthetic for insects (Cooper, 2011; Wahltinez et al., 2023), suggesting a lack of guidance from or interaction with the veterinary community may be causing entomologists to employ 'anaesthetics' that are not actually *recognized* as anaesthetics by the veterinary community. This may indicate that allowing self-regulation of welfare within entomology may be insufficient to improve animal welfare in line with modern veterinary guidance without significant changes in how these communities interact—a hypothesis in line with the judgments of professional entomologists, who have indicated that self-management is likely to be the least effective strategy for protecting insect welfare in research (Barrett et al., 2024), perhaps due to their lack of knowledge and training on insect welfare.

Importantly, the lack of *reporting* the use of a specific anaesthetic/analgesic does not necessarily represent an animal welfare issue in practice, as an anesthetic may still have been used. However, if an anaesthetic was used and not reported in the methods, this reflects a reproducibility issue as the use of different anaesthetic agents can have both behavioural and physiological

355 consequences for the animal. In *Drosophila*, CO_2 anaesthesia can delay copulation (Barron, 2000),
356 shorten lifespans (Shen et al., 2020), induces metabolic changes (Colinet & Renault, 2012), and lower
357 fecundity (MacMillan et al., 2017; Perron et al., 1972). Behavioural impacts include impaired climbing
358 and flight capacity (Bartholomew et al., 2015). Other insects are not immune, and the effects of
359 anesthesia can be wide-ranging and specific to a species' unique capacities: for example, CO_2
360 anaesthesia increased bioluminescence in a species of glow-worm (Charlton & Merritt, 2020).
361 Critically, these effects are not general to all anaesthetics, but can be a result of the specific anaesthetic
362 employed (MacMillan et al., 2017). Thus, not reporting specific use of an anaesthetic when they are
363 employed is a major concern for generating reproducible scientific research.
364 Anaesthetics may not always be needed to protect an animal's welfare during sacrifice if the
365 sacrifice method employed is instantaneous; this could mean that the low reporting of anaesthetics
366 prior to sacrifice in our studies was simply due to the common usage of instantaneous sacrifice
367 methods. However, many of the most commonly reported sacrifice methods captured in our study
368 were not instantaneous including, most notably, dissection. Dissection as a method of death made up
369 26% of our overall study population. Only 17% of papers that reported dissection as a method of death
370 reported the use of an anesthetic or analgesic prior. If this reporting accurately reflects the practices
371 employed by entomologists, vivisection (live, unanesthetized dissection of an animal) was reported in
372 83% of studies in our overall sample. This trend in reporting represents a serious public perception
373 concern: vivisection has historically been one of the oldest and most publicly inflammatory animal
374 welfare issues in research (Bates, 2017). Further, only 35% of the papers that reported the use of any
375 anesthetic used one recognized by veterinary specialists as appropriate for use prior to invasive,

surgical, or otherwise stressful/noxious procedures in insects (carbon dioxide; Cooper, 2011; Wahltinez et al., 2023).

Finally, methods of sacrifice were more frequently reported, with 60% of papers explicitly stating their sacrifice methods. However, just as the method of anaesthesia may affect reproducibility through impacts on physiology, the method of death may also impact study results. For instance, gene expression profiles change as a result of different kinds of stress - well studied, for instance, in the case of heat stress (King & Macrae, 2014). Plausibly, stress experienced by an insect when frozen instantly in liquid nitrogen is minimal; the stress of drowning (Hibicke & Nichols, 2022) in RNAlater may not be. Results from gene expression studies that collect insects using these two different methods thus may not be comparable, which can only be determined through accurate reporting of the sacrifice method employed in each study. This highlights the importance of reporting on sacrifice methods to ensure accurate comparisons across studies can be made and that results can be reproduced.

Many papers also contained a lack of detail around insect death, especially those that trap insects ("samples were collected") or extract nucleic acid ("RNA was extracted from samples following manufacturers instructions"). The vagaries introduced by use of euphemistic language (Trampe, 2017) are an ethical and practical problem that also persists in research using animals covered by legislation, as it can undermine the function of reporting methodological information with detriments to transparency, reproducibility, and reporting on ethical animal use (Taylor et al., 2024).

Trends over time and by publication

The percentage of lethal studies in our sample increased over time; those containing just invasive handling decreased. Importantly, the percentage of studies that contained an explicit ethics

statement to address animal welfare also increased over the study period (2003 to 2022) from 0% to nearly 8%. Ethics statements in general were rare, and the increase we recorded over time was found to largely be driven by policy changes at the journal *Animal Behaviour,* where ethics statements have been mandated for all studies with invertebrates at least since 2004 (Roth, personal communication).

Even with *Animal Behaviour's* policy change, the vast majority of studies from that journal after 2004 do not contain an ethics statement (> 70%). Poor reporting of welfare relevant information is common, even when studies involve vertebrate animals covered by welfare legislation (Taylor et al. 2024) and not simply journal policies. Top-down enforcement of reporting, from either administrative bodies or journals, are thus only as impactful as the research community enforcing it. Enforcement can be further complicated when both the authors and the reviewers of a community have little training in or guidance on reporting on the ethical treatment of their research animals, as is the case for entomologists (Barrett et al., 2024). Further, guidance on what to report can vary by the national origin of a publisher or a disciplinary speciality (Novak et al., 2022), resulting in confusion when authors attempt to prepare publications for submission.

Given that ethics statements are culturally non-standard for entomologists, journals considering implementing policies related to ethics reporting for insects will likely need to 1) create significant resources related to the structure and content of an ethics statement; 2) publicize the change and these resources well in advance of a mandated change (Barrett et al., 2024); and 3) engage in active enforcement of journal policies at the editorial level, not just the reviewer level, during the transition period.

Trends by taxonomic scope

Overall, journals that co-published vertebrate research contained a higher proportion of anaesthesia and sacrifice information than those journals that only publish invertebrate research. This suggests that reporting norms may vary across journals, where invertebrate-focused authors that more frequently read and publish in journals that also publish on vertebrates may be slightly more likely to consider ethics reporting even without mandates/enforcement.

Among insect orders, there were no differences in the proportion of sacrifice or anaesthesia statements prior to death. However, there were some differences in the use of anaesthetic statements prior to invasive handling. Studies with Diptera and Hymenoptera had a higher proportion of statements than Orthoptera, and studies on Diptera also had a higher proportion of statements than those on Coleoptera, Leipdoptera, and other orders. Interestingly, these results partially confirm the results of Barrett et al., (2024), where entomologists working with Hymenoptera were more likely to report the use of practices that promote insect welfare than those that did not work with Hymenoptera.

Entomologists working on Diptera and Hymenoptera do not report greater familiarity with, training on, or interest in insect welfare (Barrett et al., 2024), suggesting that these trends in reporting have not been driven by insect-welfare-related reasoning. Increased reporting of anesthetic use due to invasive handling in Hymenoptera could simply be due to occupational health and safety concerns associated with insect stings. Increased reporting in Diptera could be driven by the increased standardization of reporting when using the model organism, *Drosophila melanogaster,* in highly-controlled laboratory settings. Given that the strongest evidence for insect sentience is found in adult

Diptera, Blattodea and, to a lesser extent, Hymenoptera (Gibbons et al., 2022), it is possible reporting norms in these orders may shift more rapidly moving forward.

Finally, Hemiptera was the fifth-most common order to be studied in our dataset, representing 9% of studies. Despite their popularity as an order for research, no official reviews of the neurobiological or behavioral data related to hemipteran sentience has been conducted (Gibbons et al., 2022). Given their heavy use in entomological research, further attention to the neurobiology and behavior of hemipteran species in relation to the plausibility of their sentience is urgent to better understand our ethical obligations to these animals.

Conclusions

Despite growing concern among entomologists about insect welfare (Barrett et al., 2024), the slow increase in ethics statements over the past twenty years suggests that the field has significant room for improvement, not least to address reproducibility and translatability issues in the field. As importantly, change is likely required to address public concerns related to animal welfare (Drinkwater et al., 2019), a problem accentuated by the findings of a recent study that showed that 1) trust in scientists declines when the public learns there is an absence of oversight in invertebrate research; and 2) the public believes invertebrates should receive oversight at approximately 2/3rds the level of oversight currently afforded to vertebrates (in Canada; Brunt et al., 2022). Further, both those arguing for and skeptical of insect sentience agree that some precautionary measures are appropriate to protect welfare when using insects in research (Crump et al., 2023; Cuff et al., 2023; Drinkwater et al., 2019; Eisemann et al., 1984; Fischer & Larson, 2019; Klobučar & Fisher, 2023; Sandall & Fischer, 2019; Wigglesworth, 1980). Indeed, one of the most-cited articles *critical* of insect sentience ends with the

459 "...recommendation that insects have their nervous systems inactivated prior to traumatizing
460 manipulation...[to guard] against the remaining possibility of pain infliction" (Eisemann et al., 1984).
461 One challenge with changing the entomological community's practice is a lack of resources,
462 training, and guidance to assist community members interested in updating their practices (Barrett et
463 al., 2024). Entomologists interested in reducing animal use to the minimum number required to have
464 appropriate power can begin by looking at resources for *a priori* analyses (Lakens, 2022; Lynch & Perl,
465 2024; C. Perl, 2023). General guidance is also available from scientific societies on refining research
466 practices when using insects (Fischer et al., 2024) or animals, inclusive of invertebrates (ASAB Ethical
467 Committee/ABS Animal Care Committee, 2023); there is also general guidance on anesthesia or
468 euthanasia from veterinarians (reviewed in Barrett, 2024; and see Cooper, 2011; Wahltinez et al.,
469 2023).
470 Entomologists interested in improving their reporting practices, for welfare or reproducibility
471 reasons, should review the ARRIVE guidelines (Sert et al., 2020), guidance by Spanagel (2022), and
472 the requirements of journals like *Animal Behaviour* or *Animal Welfare*, which require more detailed
473 reporting in their ethics statements. Finally, to further facilitate entomologists' reporting of the ethical
474 use of their research animals, we have provided a table (Supplemental File 1) that can be modified and
475 included as either a supplement or in the main manuscript, to assist entomologists in capturing
476 reproducibility- and welfare-relevant data.
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Supplemental Figures

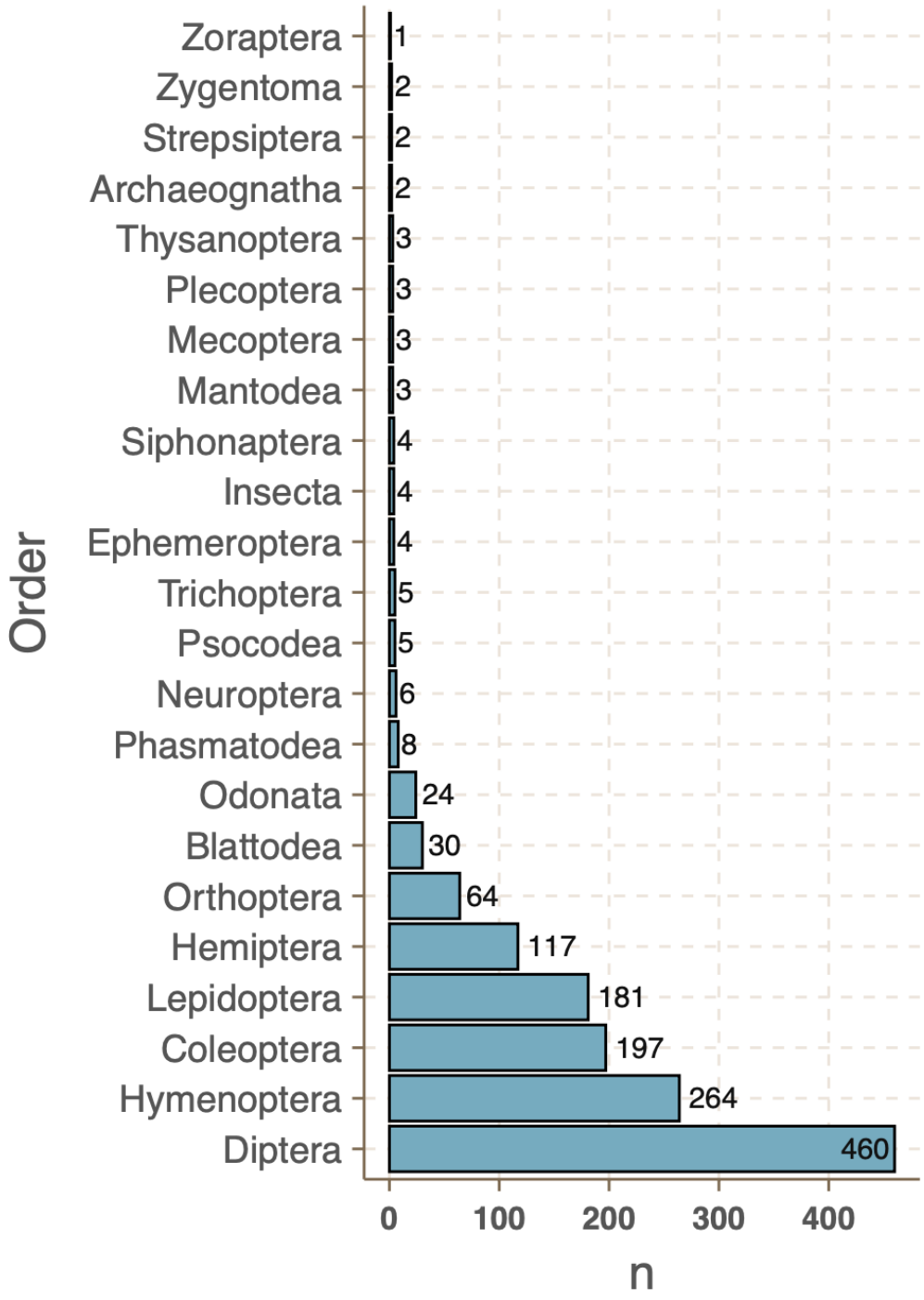
Supplemental Table 1. Experimental methods that would result in insect death but were left explicitly unstated within the sampled paper, but can be inferred from reading the methods section. Some methods are not explicit even on reading methods sections, some animals were "sacrifice" before further experimentation but the exact methods of sacrifice were not stated.

Inferred sacrifice method	n
dehydrated	1
exsanguinated	1
proteomics	1
respirometry	1
chemical compound analysis	2
protein extraction	2
supplement only	2
dissected	3
weighed	3
calcium imaging	5
euthanised	5
morphometrics	5
survival rate	6
misc. manipulation	7
nutrient content	7
trap	13
immunohistochemistry	18
microscopy	20
collected	27
electrophysiology	28
nucleic acid extraction	119

Supplemental Table 2. Anaesthetic methods employed prior to dissection (n = 304 papers).

Anaesthetic method	n
nitrogen	1
freezing	2

CO_2	18
chilling	31



Supplemental Figure 1. Number of papers that focused on insects of each order. Diptera, Hymenoptera, Coleoptera, Lepidoptera, Hemiptera, Orthoptera, and Blattodea were the most

represented orders in our analysis. Insecta accounts for papers that reported use of more than 5 orders of insects.